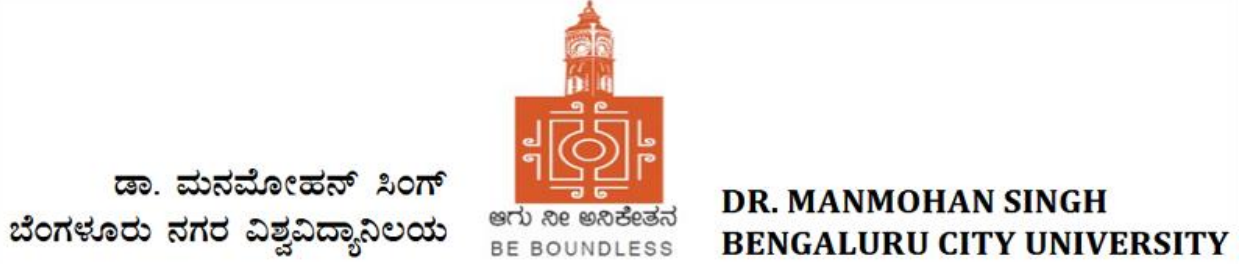


INTERNSHIP REPORT

On

“META ADS PERFORMANCE DASHBOARD”

Submitted to Dr. Manmohan Singh Bengaluru City University



In partial fulfillment for the award of BCA degree

By

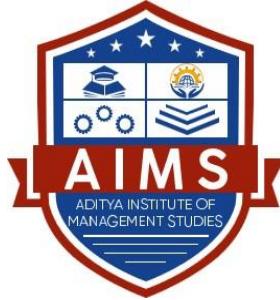
Khushboo Maurya

under the guidance of

Mrs. Yeshodha R.

Assistant professor, Department of Computer Applications

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INTERNSHIP CERTIFICATE

TO WHOMSOEVER IT MAY CONCERN

Ms. Khushboo Maurya
INTERN ID: AF05112756

This is to certify that **Khushboo Maurya**, a student of Bachelor of Computer Applications (BCA) from **Aditya Institute of Management Studies and Research**, Bangalore, has successfully completed a 3-month internship program at **Anudip Foundation**, from **12th February 2026** to **24th April 2026**.

During the internship, Candidate was actively involved in working on various projects related to Data Analysis and Business Intelligence, utilizing tools such as Power BI, Advanced Excel, and Python, MySQL. Candidate demonstrated a strong understanding of concepts, along with the ability to apply theoretical knowledge to practical business problems.

Throughout the internship period, candidate exhibited professionalism, dedication, teamwork, and a proactive attitude. Candidate shown ability to learn quickly, adapt to new challenges, and deliver quality work within deadlines was highly appreciated. Candidate also showed excellent communication and analytical skills while working on assignments and collaborating with peers.

Candidate's overall performance during the internship was found to be good.

We wish all the best for future academic and professional endeavors and we are confident that candidate will continue to achieve great success.

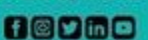


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DECLARATION OF STUDENT

I hereby declare that this Internship report on “**Meta Ads Performance Dashboard**” is an original record of my experience while conducting the internship study at Anudip foundation.

The organizational study was carried out by me, in partial fulfillment for the successful completion of VI semester BCA Course of Bengaluru City University. This Internship report has not been submitted to any other university for the award of any Degree.

PLACE: BANGALORE

DATE:

Khushboo Maurya

U18AO23S0090

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1. INTRODUCTION TO INTERNSHIP

As part of my academic curriculum in Data Analytics and Visualization, I undertook an internship focused on the design and development of a Meta Ads Performance Dashboard. This project-oriented internship provided me with valuable practical exposure to real-world data analysis and visualization techniques while enhancing my understanding of how data-driven insights can improve digital marketing strategies.

The primary objective of this internship was to analyze Meta Ads data and build interactive dashboards that effectively represent key performance indicators such as impressions, clicks, conversions, engagement, and campaign performance. The project aimed to help in understanding user behavior, evaluating campaign effectiveness, and supporting better decision-making.

Throughout the internship, I was actively involved in the complete data analysis process. The project was developed using Power BI as the primary visualization tool. Data was collected, cleaned, and transformed to ensure accuracy and consistency. Various dashboards were created to present insights in a clear and interactive manner, including user behavior analysis, campaign performance, audience insights, and overall ads performance.

The internship experience helped me gain hands-on knowledge in data cleaning, data modeling, and creating calculated measures using DAX. I worked with different types of visualizations such as bar charts, line charts, pie charts, and KPI cards to effectively communicate insights. Additionally, I learned the importance of data accuracy, consistency, and proper visualization techniques in presenting meaningful results.

Along with technical skills, this internship also played a significant role in developing my soft skills, which are equally important in a professional environment. I improved my communication skills by clearly expressing ideas and explaining dashboard insights during discussions. I actively collaborated with peers, which enhanced my teamwork and coordination abilities. While working on the project, I strengthened my problem-solving skills by identifying data-related issues and resolving them effectively.

I also developed strong time management skills by completing tasks within deadlines and maintaining consistency throughout the internship.

My presentation skills improved as I confidently presented my dashboards and explained key findings in a structured manner. Overall, this experience contributed to my professional growth by building confidence, adaptability, and a disciplined approach towards work.

This internship served as a bridge between theoretical knowledge and practical implementation. It enhanced my analytical thinking, problem-solving skills, and understanding of business intelligence concepts. By the end of the internship, I developed a deeper understanding of how data analytics can be used to optimize advertising strategies and improve performance in digital marketing.

This internship has equipped me with essential skills in data analysis and visualization, preparing me for future roles in the field of data analytics and business intelligence.

1.1 MOTIVATION

The motivation behind this project is to understand how data analytics can be used to improve digital marketing performance, especially in the context of Meta Ads. In today's competitive digital environment, businesses rely heavily on online advertising to reach their target audience, increase engagement, and drive conversions. However, without proper analysis, it becomes difficult to measure the effectiveness of these campaigns.

This project was undertaken with the aim of gaining practical knowledge in analyzing advertising data and converting it into meaningful insights. By working on real-world datasets, I was motivated to explore how different metrics such as impressions, clicks, conversions, and engagement rates influence the success of advertising campaigns.

Another key motivation was to develop strong skills in data visualization using tools like Power BI. Creating interactive dashboards helped in presenting complex data in a simple and understandable manner, enabling better decision-making. It also encouraged me to think analytically and identify patterns and trends in the data.

In addition to technical learning, this project also motivated me to improve my soft skills. I enhanced my communication skills by clearly explaining data insights and presenting dashboard results effectively. Working on the project helped me develop teamwork and collaboration skills through discussions and idea sharing. I also strengthened my problem-solving abilities by

identifying data issues and resolving them systematically. Furthermore, I improved my time management skills by organizing tasks efficiently and meeting deadlines. My presentation skills were enhanced as I confidently explained my findings during reviews. Overall, this experience contributed to my professional growth by building confidence, adaptability, and a structured approach towards work.

Overall, this project motivated me to enhance both my technical and analytical skills while understanding the importance of data-driven strategies in modern digital marketing.

1.2 INTERNSHIP OBJECTIVES & GOALS

The main objective of this internship is to gain practical knowledge in data analysis and visualization by working on a real-world project based on Meta Ads performance. The internship focuses on understanding how advertising data can be analyzed to evaluate campaign effectiveness and support better decision-making.

The specific objectives of this internship are:

- To collect, clean, and transform raw data into a structured format suitable for analysis.
- To analyze key performance metrics such as impressions, clicks, conversions, and engagement.
- To design and develop interactive dashboards using Power BI for effective data visualization.
- To use DAX formulas to create calculated measures and derive meaningful insights.
- To understand user behavior and campaign performance through data analysis.
- To improve analytical thinking and problem-solving skills in a real-world scenario.

The internship aims to bridge the gap between theoretical knowledge and practical application by providing hands-on experience in business intelligence and data visualization.

1.3 SCOPE AND COVERAGE OF THE INTERNSHIP REPORT

The scope of this internship report is focused on analyzing and visualizing Meta Ads data to understand advertising performance and user engagement. The report covers the complete process of transforming raw data into meaningful insights using data analytics and visualization techniques, providing a structured approach to data-driven decision-making.

This internship involves working with datasets related to digital advertising, including key metrics such as impressions, clicks, conversions, engagement, reach, and campaign details. The scope includes data collection, data cleaning, data transformation, and data modeling, followed by the creation of interactive dashboards using Power BI to present the analyzed results in a clear, concise, and user-friendly manner.

The report also highlights various aspects of data analysis, such as identifying trends, tracking performance over time, comparing campaign effectiveness, and understanding audience behavior. Advanced visualization techniques, including the use of charts, graphs, KPI indicators, slicers, and filters, are applied to enhance interactivity and improve data interpretation. Multiple dashboards have been developed to represent key insights, including user behavior analysis, campaign performance, audience insights, engagement quality, and overall, Meta Ads performance.

In addition to technical aspects, the scope of this internship also includes the development of essential soft skills. Throughout the project, I enhanced my communication skills by effectively explaining data insights and participating in discussions. I improved my teamwork and collaboration abilities while working with peers and sharing ideas. Handling data-related challenges helped strengthen my problem-solving and analytical thinking skills.

Furthermore, I developed strong time management skills by organizing tasks efficiently, prioritizing activities, and completing them within deadlines. My presentation skills improved as I confidently presented dashboards and explained key findings in a structured and professional manner. Overall, the internship contributed to my professional growth by improving confidence, adaptability, and a disciplined approach to work.

The scope also briefly covers the understanding of concepts such as API integration, real-time data usage, and the role of databases in managing large datasets, which are important for real-world implementation of such dashboards.

However, the scope of this project is limited to data visualization and analysis only. It does not include the creation, execution, or management of actual advertising campaigns. The findings are based on the available dataset and are intended primarily for learning, analysis, and demonstration purposes.

Overall, this internship report provides a comprehensive understanding of how data analytics and visualization can be applied in digital marketing to support better decision-making, improve campaign performance, and enhance business strategies.

2. INTRODUCTION TO COMPANY



Anudip Foundation is a non-profit organization established in 2007 with the aim of creating sustainable livelihood opportunities for underprivileged communities through skill development and digital training. Headquartered in Kolkata, India, the organization focuses on empowering youth and women from marginalized backgrounds by providing them with industry-relevant skills and employment opportunities.

The foundation was founded by social entrepreneurs with a vision to bridge the gap between unemployment and skill-based education. Over the years, it has grown significantly and now operates multiple training centers across India, offering programs in information technology, digital skills, and professional development.

Anudip Foundation works closely with various corporate partners and organizations such as Accenture, Capgemini, Amazon, and Infosys to ensure that its training programs are aligned with current industry requirements. These collaborations help in providing placement opportunities and career support to learners after completion of their training.

The organization has made a significant social impact by transforming the lives of more than 500,000 individuals through its training initiatives. It focuses on equipping learners with both technical and soft skills, enabling them to secure sustainable employment and improve their standard of living.

Anudip Foundation's mission is to empower disadvantaged youth with market-driven skills and create digital livelihoods, while its vision is to promote inclusive growth and reduce inequality through education and employment opportunities.

Anudip Foundation plays a crucial role in bridging the digital divide and supporting economic development by preparing individuals for the modern workforce through technology-driven education and skill development programs.

2.1 KEY LEARNING EXPERIENCE & INDUSTRY EXPOSURE

The internship at Anudip Foundation offered a comprehensive and practical learning experience by combining technical knowledge with real-world application. The key aspects of the internship experience are as follows:

Exposure to Real World Projects:

I worked on a live project titled **Meta Ads Performance Dashboard**, which helped me understand real-world data analysis processes, dashboard development, and business-oriented decision-making.

Hands-on-Learning:

The internship provided practical exposure to industry-standard tools such as Power BI, Excel, and DAX. I gained experience in data cleaning, transformation, modeling, and visualization, which enhanced my technical capabilities.

Collaboration with Industry Experts:

I received guidance from experienced mentors and professionals who provided valuable insights into real-world challenges, best practices, and industry expectations. This interaction helped me align my work with professional standards.

Operational Understanding:

I gained knowledge about the structure and workflow of training programs, understanding how learning modules are designed to meet industry requirements and enhance skill development.

Student Interaction and Support:

I actively participated in discussions, collaborative sessions, and knowledge-sharing activities. This improved my communication skills and helped me effectively interact with peers and mentors while resolving queries related to the project.

Innovation and Improvement:

I contributed to enhancing dashboard design and improving data visualization techniques to make

insights more clear, interactive, and user-friendly. This encouraged creativity and analytical thinking.

Marketing and Analytical Exposure:

I developed an understanding of how data analytics supports digital marketing. By analyzing Meta Ads data, I learned how campaign performance can be evaluated and optimized using data-driven strategies.

Soft Skills Development:

In addition to technical learning, the internship significantly improved my soft skills. I enhanced my communication, teamwork, problem-solving, and time management abilities. I also developed confidence in presenting insights and working in a structured and professional manner.

These aspects collectively ensured a smooth and enriching learning experience, covering all stages from data handling and analysis to dashboard development and presentation. The internship at Anudip Foundation provided a strong foundation in both technical and professional skills, preparing me for real-world industry challenges.

3. PROJECT OVERVIEW: META ADS PERFORMANCE DASHBOARD

The Meta Ads Performance Dashboard project began with understanding the dataset, identifying key metrics, and defining the objectives of the analysis. As part of the internship, I focused on applying my data analytics and visualization skills to design and develop an interactive and user-friendly dashboard. This involved preparing the data, planning the dashboard layout, and identifying important performance indicators.

Using tools like Power BI for visualization, I worked on developing dashboards that represent key metrics such as impressions, clicks, conversions, engagement, and campaign performance. Special emphasis was placed on creating clear and interactive visualizations, including KPI cards, bar charts, line charts, and donut charts, to ensure effective data representation and easy interpretation. The project also included data cleaning, transformation, and the use of DAX formulas to create calculated measures such as Click-Through Rate (CTR) and Conversion Rate. Slicers and filters were implemented to allow dynamic interaction with the data, enabling users to explore insights based on different parameters like campaign, date, and audience segments. The goal of this project was to analyze advertising data and provide meaningful insights that can help improve campaign performance and decision-making. The dashboard allows users to easily monitor key metrics, compare campaign effectiveness, and understand user behavior, ultimately contributing to better digital marketing strategies.

3.1 PROJECT DESCRIPTION & FUNCTIONAL OVERVIEW

The Meta Ads Performance Dashboard is a comprehensive data visualization solution developed during my Data Analytics internship. It is designed to simplify and analyze digital advertising performance by transforming raw data into meaningful insights. The dashboard focuses on key metrics such as impressions, clicks, conversions, engagement, and campaign performance, enabling better decision-making in digital marketing. The project began with in-depth analysis of the dataset and understanding the requirements of digital marketing analytics. By studying different advertising metrics and their impact, I developed a structured plan to create dashboards that are interactive, user-friendly, and informative. The goal was to ensure clarity in data representation and provide actionable insights for improving campaign performance.

The dashboard was developed using Power BI, which provided a powerful platform for data visualization and analysis. Various visual elements such as KPI cards, bar charts, line charts, pie charts, and scatter plots were used to represent data effectively. The interface was designed to be clean, responsive, and easy to navigate, allowing users to quickly understand key insights.

Data preparation was an important part of the project. The dataset was cleaned and transformed to remove inconsistencies and ensure accuracy. DAX (Data Analysis Expressions) was used to create calculated measures such as Click-Through Rate (CTR), Conversion Rate, and Total Engagement. These measures helped in deriving deeper insights from the data.

The project includes several important dashboard modules:

- **Overview Dashboard:** Provides a summary of key metrics such as total impressions, clicks, conversions, and overall performance.
- **User Behavior Analysis:** Focuses on understanding how users interact with ads, including activity patterns and engagement trends.
- **Campaign Performance Dashboard:** Compares different campaigns based on performance metrics to identify the most effective strategies.
- **Audience Insights Dashboard:** Analyzes demographic data such as age group and user segments to understand the target audience better.
- **Engagement Quality Dashboard:** Evaluates the quality of engagement by analyzing user interactions and conversion effectiveness.
- **Meta Ads Performance Dashboard:** Presents a complete overview of advertising performance, including trends, ROI analysis, and campaign comparisons.
- **Data Interaction Features:** Slicers and filters were implemented to allow dynamic analysis based on campaign, date, and audience parameters.

Throughout the project, emphasis was placed on data accuracy, visualization clarity, and user interaction. This project enhanced my skills in data analysis, dashboard design, and business intelligence concepts, while also improving my ability to interpret data and present insights effectively.

3.2 TECHNOLOGIES AND TOOLS UTILIZED

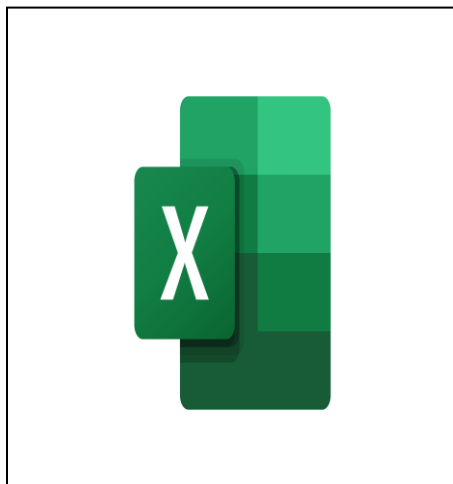


The primary tool used in this project is **Power BI**, which was utilized for data visualization and dashboard creation. Power BI provides powerful features such as interactive charts, KPI cards, slicers, and filters, enabling users to explore data easily and gain meaningful insights. For data handling and preprocessing, Microsoft Excel was used. It helped in organizing the dataset, performing initial cleaning, and preparing the data before importing it into Power BI.



DAX (Data Analysis Expressions) was used within Power BI to create calculated measures and perform advanced data analysis. Key performance metrics such as Click-Through Rate (CTR), Conversion Rate, Total Engagement, and other derived values were calculated using DAX formulas.

The project also involved data modeling techniques, where relationships between different data fields were established to ensure accurate analysis and reporting.



Microsoft Excel is a widely used spreadsheet application for data organization, analysis, and manipulation. It was used in this project to store and preprocess IPL data before importing it into Power BI. Excel helped in cleaning the data, formatting columns, and ensuring consistency. Overall, the technologies used in this project ensured efficient data processing, accurate analysis and interactive visualization, making it easier to understand Meta Ads performance and support data-driven decision-making.

3.3 FRONTEND DESIGN & DATA VISUALIZATION

To develop an interactive and user-friendly interface for data visualization, I used:

- **Power BI**

Power BI is a powerful data visualization tool developed by Microsoft, ideal for creating interactive dashboards and reports. It allows efficient representation and real-time updating of data through dynamic visuals. In this project, Power BI was used to create multiple dashboards such as campaign performance, user behavior analysis, audience insights, and overall, Meta Ads performance, ensuring a smooth and interactive user experience.

- **Data Visualization Techniques**

Data Visualization Techniques include the use of charts such as bar charts, line charts, pie charts, donut charts, and KPI cards. These visuals help in presenting complex data in a simple and understandable format. They allow users to easily identify trends, compare performance, and gain insights from the data.

- **DAX (Data Analysis Expressions)**

DAX (Data Analysis Expressions) is used to create calculated measures and perform advanced data analysis. It helps in deriving important metrics such as Click-Through Rate (CTR), Conversion Rate, and Total Engagement, which are essential for evaluating ad performance.

Additionally, interactive features like slicers and filters were implemented to allow users to explore the data dynamically based on different parameters such as campaign, date, and audience.

3.4 BACKEND PROCESSING & DATA MODELING

The backend of the Meta Ads Performance Dashboard focuses on data processing, transformation, and modeling to ensure accurate and meaningful analysis. Unlike traditional application development, the backend in this project is handled within Power BI and Excel.

The first step involved importing the dataset into Power BI. Before visualization, the data was cleaned and transformed using Power Query Editor. This process included removing duplicate values, handling missing data, correcting data types, and organizing the dataset into a structured format suitable for analysis.

Data modeling was performed to establish relationships between different data fields such as campaigns, dates, user details, and performance metrics. Proper relationships ensured accurate calculations and efficient data flow across different dashboards.

DAX (Data Analysis Expressions) was used extensively to create calculated measures and perform advanced analysis. Important metrics such as Total Impressions, Total Clicks, CTR (Click-Through Rate), Conversion Rate, and Engagement were calculated using DAX formulas.

The backend also ensures that all visualizations are dynamically connected to the data. Any changes made through filters or slicers are processed instantly, providing real-time insights to the user.

Overall, the backend plays a crucial role in transforming raw data into structured information, enabling accurate analysis and supporting interactive dashboard functionality.

3.5 DATA MANAGEMENT & DATASET STRUCTURE

The database plays a crucial role in storing, managing, and retrieving data efficiently for analysis and visualization. In the Meta Ads Performance Dashboard project, the dataset acts as the primary source of information used to generate insights and build interactive dashboards.

For this project, the data was sourced from structured files such as Excel/CSV, which contained information related to advertising campaigns. The dataset included key attributes such as campaign name, impressions, clicks, conversions, engagement metrics, cost, and date-wise performance.

The database structure was designed in a tabular format, where each row represents a record of advertising data and each column represents specific attributes. This structured format made it easier to perform data cleaning, transformation, and analysis using tools like Power BI.

Data preprocessing was an important step in managing the database. It involved handling missing values, removing duplicates, correcting inconsistencies, and ensuring data accuracy. These steps helped in improving the quality and reliability of the data used for analysis.

In Power BI, the data was further organized using data modeling techniques. Relationships were established between different tables (if applicable), and calculated columns and measures were created using DAX to enhance data analysis.

Although this project used a static dataset, in real-world scenarios, databases such as MySQL, SQL Server, or cloud-based databases are commonly used to store and manage large volumes of data.

These databases can be connected directly to Power BI for real-time data updates and dynamic reporting.

Overall, the database component of this project ensured that data was well-organized, accurate, and ready for analysis, forming the foundation for creating meaningful insights and effective visualizations.

ABOUT THE DATA

The dataset used in this project represents Meta Ads Performance Data, which includes information related to advertising campaigns, ads, user demographics, and user interaction events. It is structured to reflect how platforms like Facebook and Instagram (Meta) capture and store advertising data.

The primary purpose of this dataset is to analyze advertising performance, understand user engagement, and evaluate campaign effectiveness using key performance indicators (KPIs). These KPIs include impressions (how often ads are viewed), clicks (user interaction with ads), conversions (purchases or desired actions), and performance metrics such as Cost Per Click (CPC), Cost Per Mille (CPM), Click-Through Rate (CTR), and Return on Ad Spend (ROAS).

The dataset also provides valuable insights into audience behavior, including demographics such as age, gender, location, and user interests. This helps in understanding how different audience segments respond to advertisements and supports better targeting strategies.

This dataset is highly suitable for building an interactive Power BI dashboard to monitor campaign performance, analyze user behavior, track budget utilization, and generate meaningful insights for data-driven decision-making.

USE OF EACH TABLE

Table 1: Ad Events

This table contains event-level data such as impressions, clicks, and conversions. It acts as the primary (fact) table in the data model, as most KPIs are derived from these events. It is used to analyze how users interact with ads over time and measure engagement levels.

event_id	ad_id	user_id	timestamp	day_of_week	time_of_day	event_type	Event Date	Event Hour
817	84	43f91	5/29/2025 12:35:00 AM	Thursday	Night	Impression	Thursday, May 29, 2025	0
11847	84	97551	7/3/2025 5:45:00 AM	Thursday	Night	Impression	Thursday, July 3, 2025	5
16894	84	4bd90	6/26/2025 12:53:00 AM	Thursday	Night	Impression	Thursday, June 26, 2025	0
18721	84	c61da	7/17/2025 3:01:00 AM	Thursday	Night	Impression	Thursday, July 17, 2025	3
24761	84	bb595	5/8/2025 12:47:00 AM	Thursday	Night	Impression	Thursday, May 8, 2025	0
25537	84	d4330	7/17/2025 12:52:00 AM	Thursday	Night	Impression	Thursday, July 17, 2025	0
38799	84	0ca44	7/10/2025 3:21:00 AM	Thursday	Night	Impression	Thursday, July 10, 2025	3
42634	84	77d4b	7/3/2025 5:15:00 AM	Thursday	Night	Impression	Thursday, July 3, 2025	5
47398	84	1bbd0	7/24/2025 2:44:00 AM	Thursday	Night	Impression	Thursday, July 24, 2025	2
49823	84	ed67e	7/31/2025 3:39:00 AM	Thursday	Night	Impression	Thursday, July 31, 2025	3
58129	84	999f6	5/22/2025 5:05:00 AM	Thursday	Night	Impression	Thursday, May 22, 2025	5
67160	84	1f39f	7/3/2025 5:02:00 AM	Thursday	Night	Impression	Thursday, July 3, 2025	5
72906	84	a7c22	7/10/2025 12:02:00 AM	Thursday	Night	Impression	Thursday, July 10, 2025	0
76826	84	0fe7a	7/3/2025 2:30:00 AM	Thursday	Night	Impression	Thursday, July 3, 2025	2
92244	84	a0f2a	7/10/2025 3:56:00 AM	Thursday	Night	Impression	Thursday, July 10, 2025	3
92954	84	02f27	7/31/2025 5:55:00 AM	Thursday	Night	Impression	Thursday, July 31, 2025	5
95048	84	c22b3	7/31/2025 12:15:00 AM	Thursday	Night	Impression	Thursday, July 31, 2025	0
101995	84	0a8dc	6/12/2025 3:12:00 AM	Thursday	Night	Impression	Thursday, June 12, 2025	3
102592	84	4f8cc	5/8/2025 4:22:00 AM	Thursday	Night	Impression	Thursday, May 8, 2025	4
121930	84	ec5d8	6/26/2025 4:42:00 AM	Thursday	Night	Impression	Thursday, June 26, 2025	4
122465	84	692b3	6/5/2025 1:07:00 AM	Thursday	Night	Impression	Thursday, June 5, 2025	1
122942	84	9cd1d	7/3/2025 5:24:00 AM	Thursday	Night	Impression	Thursday, July 3, 2025	5
125611	84	513ef	7/10/2025 3:40:00 AM	Thursday	Night	Impression	Thursday, July 10, 2025	3
129500	84	80Rha	7/24/2025 5:15:00 AM	Thursday	Night	Impression	Thursday, July 24, 2025	5

Table: ad_events data set 1 (400,000 rows)

Table 2: Ads

This table provides detailed information about individual ads, including ad type, creative format, and platform. It helps in analyzing performance at the ad level, such as comparing different ad creatives (e.g., video ads vs image ads) and understanding which type performs better.

ad_id	campaign_id	ad_platform	ad_type	target_gender	target_age_group	target_interests
1	28	Facebook	Video	Female	35-44	art, technology
4	28	Facebook	Stories	Female	25-34	news
5	24	Instagram	Image	Female	25-34	news
7	9	Facebook	Image	Female	18-24	fashion, gaming
10	17	Facebook	Image	Female	18-24	art
12	45	Facebook	Stories	Female	35-44	gaming, technology
13	6	Facebook	Stories	Female	All	health, food
14	21	Instagram	Image	Female	35-44	finance, lifestyle
16	18	Facebook	Video	Female	35-44	food, gaming
19	3	Facebook	Video	Female	18-24	gaming
24	23	Facebook	Video	Female	All	technology
26	42	Facebook	Image	Female	All	gaming, health
27	4	Instagram	Video	Female	All	health, fashion
28	48	Facebook	Image	Female	25-34	finance, travel
30	49	Facebook	Carousel	Female	25-34	sports
31	21	Facebook	Image	Female	25-34	fashion
33	15	Instagram	Carousel	Female	All	photography
34	26	Instagram	Stories	Female	35-44	sports, technology
35	18	Instagram	Image	Female	25-34	travel, finance
42	19	Facebook	Video	Female	25-34	health
44	40	Facebook	Video	Female	18-24	fashion, gaming
45	38	Facebook	Carousel	Female	All	sports, gaming
47	17	Instagram	Image	Female	35-44	fitness, sports
50	5	Instagram	Stories	Female	35-44	news

Table: ads data set (200 rows)

Table 3: Campaigns

This table contains high-level information about advertising campaigns, including campaign name, duration, and budget allocation. It is used to evaluate overall campaign performance and calculate cost-related KPIs such as CPC, CPM, and ROAS.

campaign_id	name	start_date	end_date	duration_days	total_budget
1	Campaign_1_Launch	25-05-2025	23-07-2025	59	24021.32
2	Campaign_2_Launch	16-04-2025	7/7/2025	82	79342.41
3	Campaign_3_Winter	4/5/2025	29-06-2025	56	14343.25
4	Campaign_4_Summer	4/6/2025	8/8/2025	65	45326.6
5	Campaign_5_Launch	11/7/2025	28-08-2025	48	68376.69
6	Campaign_6_Winter	21-06-2025	13-09-2025	84	78607.49
7	Campaign_7_Winter	25-05-2025	5/8/2025	72	43744.59
8	Campaign_8_Q3	25-02-2025	7/4/2025	41	39953.19
9	Campaign_9_Launch	25-05-2025	13-07-2025	49	40094.07
10	Campaign_10_Winter	17-05-2025	21-07-2025	65	19669.27
11	Campaign_11_Q3	24-04-2025	2/7/2025	69	79067.91
12	Campaign_12_Q3	8/3/2025	1/6/2025	85	30525.65
13	Campaign_13_Winter	19-04-2025	1/7/2025	73	21855.42
14	Campaign_14_Summer	15-04-2025	4/6/2025	50	39849.94
15	Campaign_15_Launch	21-03-2025	11/6/2025	82	85407.23
16	Campaign_16_Winter	7/3/2025	4/6/2025	89	71521.88
17	Campaign_17_Launch	22-04-2025	18-07-2025	87	86675.92
18	Campaign_18_Q3	19-04-2025	15-06-2025	57	48832.61
19	Campaign_19_Winter	6/3/2025	15-05-2025	70	33182.45
20	Campaign_20_Winter	17-02-2025	18-05-2025	90	98904.66
21	Campaign_21_Winter	19-04-2025	22-06-2025	64	37290.81
22	Campaign_22_Q3	17-03-2025	22-04-2025	36	35743.4
23	Campaign_23_Winter	10/7/2025	2/10/2025	84	26001.67
24	Campaign_24_Summer	30-03-2025	23-05-2025	54	56602.87

Table: campaigns data set (50 rows)

Table 4: Users

This table includes demographic details of users such as age, gender, location, and interests. It is used for audience segmentation and helps in analyzing whether ads are reaching the intended target audience effectively.

user_id	user_gender	user_age	age_group	country	location	interests
5341	Male	26	25-34	United States	Lake Angelamouth	travel, news
18898	Male	26	25-34	United States	Jadefort	travel
27077	Male	26	25-34	United States	Nicholasport	gaming
27958	Male	26	25-34	United States	Romanmouth	fashion, health
31381	Male	26	25-34	United States	Paulmouth	travel
67468	Male	26	25-34	United States	Johnsonburgh	finance
71097	Male	26	25-34	United States	East Joshuamouth	news, art, technology
77938	Male	26	25-34	United States	Wongfurt	technology, photography, finance
78875	Male	26	25-34	United States	Hensleyberg	food, lifestyle
80203	Male	26	25-34	United States	Pamelaville	sports, travel, finance
88305	Male	26	25-34	United States	Ericamouth	sports, photography, art
94490	Male	26	25-34	United States	Kristintown	sports, news
94660	Male	26	25-34	United States	West Kristintown	fashion
95882	Male	26	25-34	United States	Lake Aprilbury	food, health
4.00E+226	Male	26	25-34	United States	Dylanhaven	finance, photography, fitness
0162f	Male	26	25-34	United States	Phillipsmouth	fitness
03a52	Male	26	25-34	United States	Jensenside	art
0429c	Male	26	25-34	United States	East Natashabury	fitness, sports
0458d	Male	26	25-34	United States	South Toddport	finance, travel
0e9cb	Male	26	25-34	United States	North John	lifestyle, photography, fitness
0e9ed	Male	26	25-34	United States	West Deborah	fitness, news, health
116eb	Male	26	25-34	United States	New Williamchester	technology
12b06	Male	26	25-34	United States	Jonesville	technology
147a0	Male	26	25-34	United States	Thorntonland	fitness, health

Table: users data set (9,841 rows)

3.6 ANALYTICAL LOGIC (DAX) & KPI DEFINITIONS

The Meta Ads Performance Dashboard is developed using Power BI, where the core logic is implemented using DAX (Data Analysis Expressions). Instead of traditional programming code, this project uses calculated measures and data modeling techniques to analyze and visualize data effectively.

KPIs & DEFINITIONS

KPI	Definition	Formula	Example Use
Impressions	Number of times ads were displayed.	Count of event_type = Impression	Measure reach
Clicks	Number of times users clicked ads.	Count of event_type = Click	Measure engagement intent
Shares	Number of times ads were shared.	Count of event_type = Share	Viral engagement
Comments	Number of user comments on ads.	Count of event_type = Comment	User sentiment & feedback
Purchases	Number of purchases made after seeing ads.	Count of event_type = Purchase	Conversions
Engagements	Total interactions (Clicks+ Shares + Comments).	Clicks + Shares+ Comments	Engagement
CTR (Click Through Rate)	% of impressions that resulted in clicks.	$(\text{Clicks} \div \text{Impressions}) \times 100$	Ad effectiveness
Engagement Rate	% of impressions that resulted in engagements.	$(\text{Engagements} \div \text{Impressions}) \times 100$	Overall ad appeal
Conversion Rate	% of clicks that resulted in purchases.	$(\text{Purchases} \div \text{Clicks}) \times 100$	Funnel efficiency
Purchase Rate	% of impressions that resulted in purchases.	$(\text{Purchases} \div \text{Impressions}) \times 100$	Conversion from reach
Total Budget	Total spend allocated to campaigns.	Sum of campaigns. total_budget	Cost analysis
Avg. Budget per Campaign	Average budget allocation per campaign.	Total Budget $\div$ Campaign Count	Budget distribution

KPI Metrics

- **Impressions:** 216K: Total times the ads were shown. Good reach.
- **Clicks:** 25.4K: Number of people who clicked on the ads.
- **Shares:** 1.3K, **Comments:** 2.6K: Indicators of organic engagement (beyond paid reach).
- **Purchases (Conversions):** 1.3K: Real customer acquisitions from ads.
- **Engagements:** 29K: Sum of clicks, likes, shares, comments.
- **CTR (Click-Through Rate):** 11.76%: Strong performance (above industry average ~1-2%). Ads are very attractive.
- **Engagement Rate:** 13.56%: Very healthy; content resonates with the audience.
- **Conversion Rate:** 5.21%: Out of all clicks, 5.21% converted into purchases. Good but could improve with landing page optimization.
- **Purchase Rate:** 0.61%: Out of impressions, only 0.61% resulted in purchases. Low conversion funnel efficiency (room to optimize).
- **Total Budget:** 2.5M: Total ad spends.
- **Avg Budget per Campaign:** 50.7K: Suggests multiple campaigns were run.
- **Insight:** Ads are performing strongly in visibility and engagement, but actual purchase efficiency is weak: need to optimize targeting/landing pages.
- **High CTR (11.76%) and Engagement Rate (13.56%)** → clearly indicate that the ad creatives, messaging, and targeting at the top of the funnel are very effective.
People are interested enough to click, like, share, or comment.
- **Low Purchase Rate (0.61%) and only 1.3K conversions out of 216K impressions** shows a sharp drop-off in the lower funnel. This is a classic case of "awareness and interest" being strong but "action (purchase)" being weak.

Engagement Breakdown

- **By Gender** (Donut Chart)

- o **Female:** 13K (43%)

- o **Male:** 6K (22%)

- o **Other/Not Specified:** 10K (35%)

Females engage more than males; campaigns could be tailored toward female audiences.

- **By Target Age** (Bar Chart)

- o **Peak engagement:** 20–30 age group (especially early 20s).

- o Drops significantly after 35+.

Primary audience = Young adults.

Insight: Target ads towards females aged 18–30 for better ROI.

Geographic Distribution

- **Top Engaged Countries**

- o US, India, Brazil, Germany, UK are major contributors.

Insight: Focus campaigns in India & US (high potential, high engagement), and premium campaigns in Germany/UK (better conversion potential due to higher purchasing power).

Time-Based Trends

- **Weekly Engagement Trend** (Stacked Bar)

- o Fairly consistent across weeks, with no sharp drop.

- o Steady engagement shows ads maintain attention.

- **Hourly Engagement Trend** (Line Chart)

- o Peaks around late afternoon & evening (~15–20 hours).

o Lowest engagement early morning (~0–5 hours).

Insight: Schedule ad delivery during afternoons & evenings for maximum impact.

Calendar View

- Engagements are mapped to days in June.
- Certain dates (like 19th–21st, 25th–27th) show higher highlights.

Campaign activity peaks on specific days, possibly due to launches/promotions.

Insight: Weekly promotions/events significantly drive engagement.

Analysis by Ad Type (Bottom-Right Table)

Video ads have the highest CTR, CR, ER (best-performing).

Stories ads also perform strongly with higher impressions.

Images/Carousels have decent performance but slightly lower conversions.

Insight: Focus budget more on Video & Story ads for better ROI.

Ad Type	Impressions	Clicks	CTR	Purchase Rate	Conversion Rate	Engagement Rate
Carousel	48K	6K	11.7%	0.59%	5.1%	13.4%
Image	51K	6K	11.7%	0.57%	4.9%	13.5%
Stories	72K	8K	11.8%	0.65%	5.2%	13.6%
Video	46K	5K	11.9%	0.62%	5.2%	13.7%

DASHBOARD OUTPUTS & VISUAL INSIGHTS

1.This dashboard summarizes Meta ad performance, showing key metrics like impressions, clicks, purchases, and conversion rate. It also provides insights into audience demographics, engagement trends, and campaign effectiveness to support better decision-making.

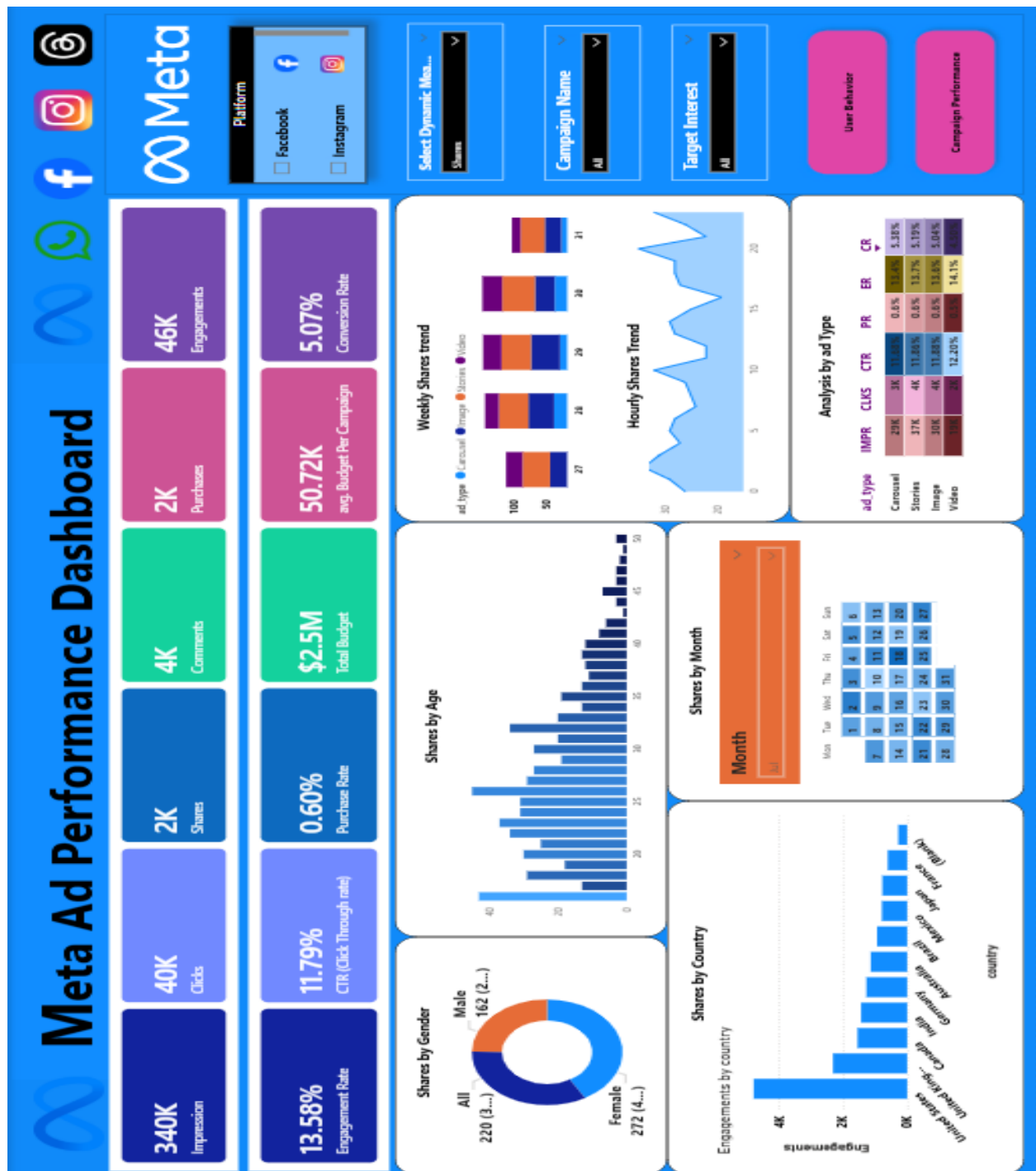


Figure 1.0

2.This dashboard summarizes Meta ad performance, showing key metrics like impressions, clicks, purchases, and conversion rate. It also provides insights into audience demographics, engagement trends, and campaign effectiveness to support better decision-making.

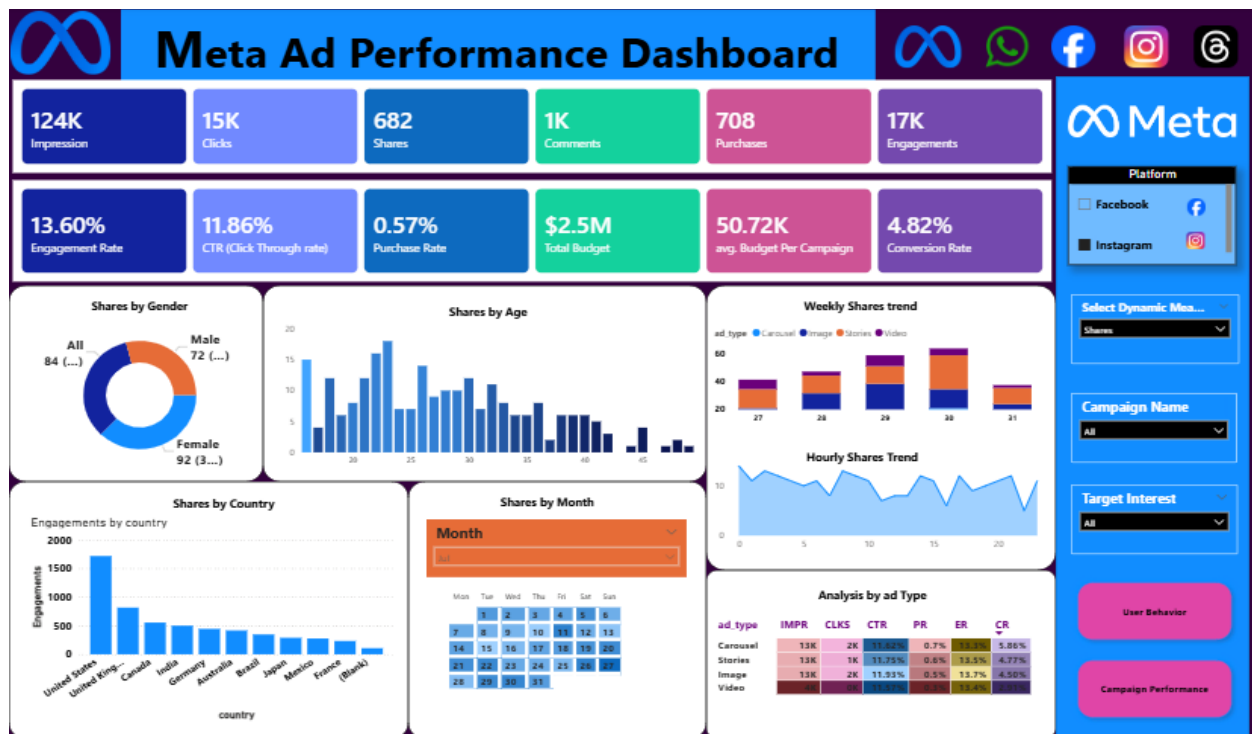
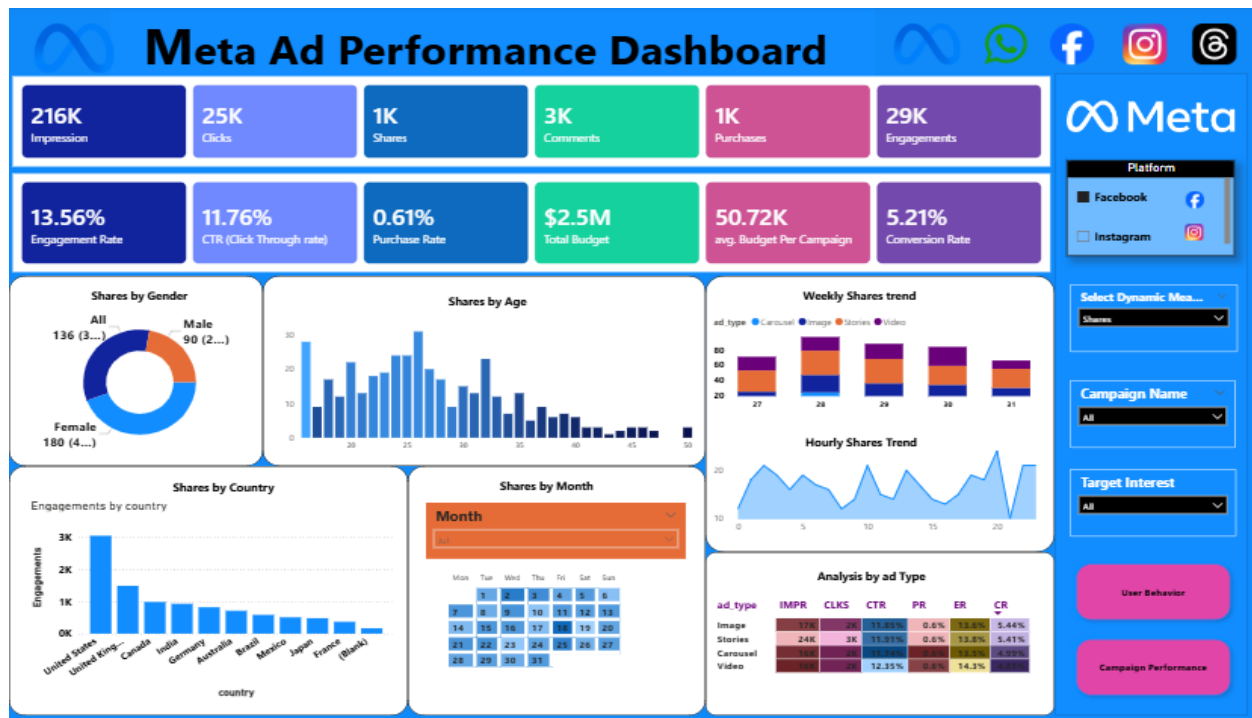


Figure 1.1

3. This dataset analyzes user behavior on Meta platforms like Facebook and Instagram. It shows which age groups are most active and engaging. It identifies the peak time when users are most active. It tracks device usage (mobile or desktop). It helps understand how users interact with content to improve targeting.

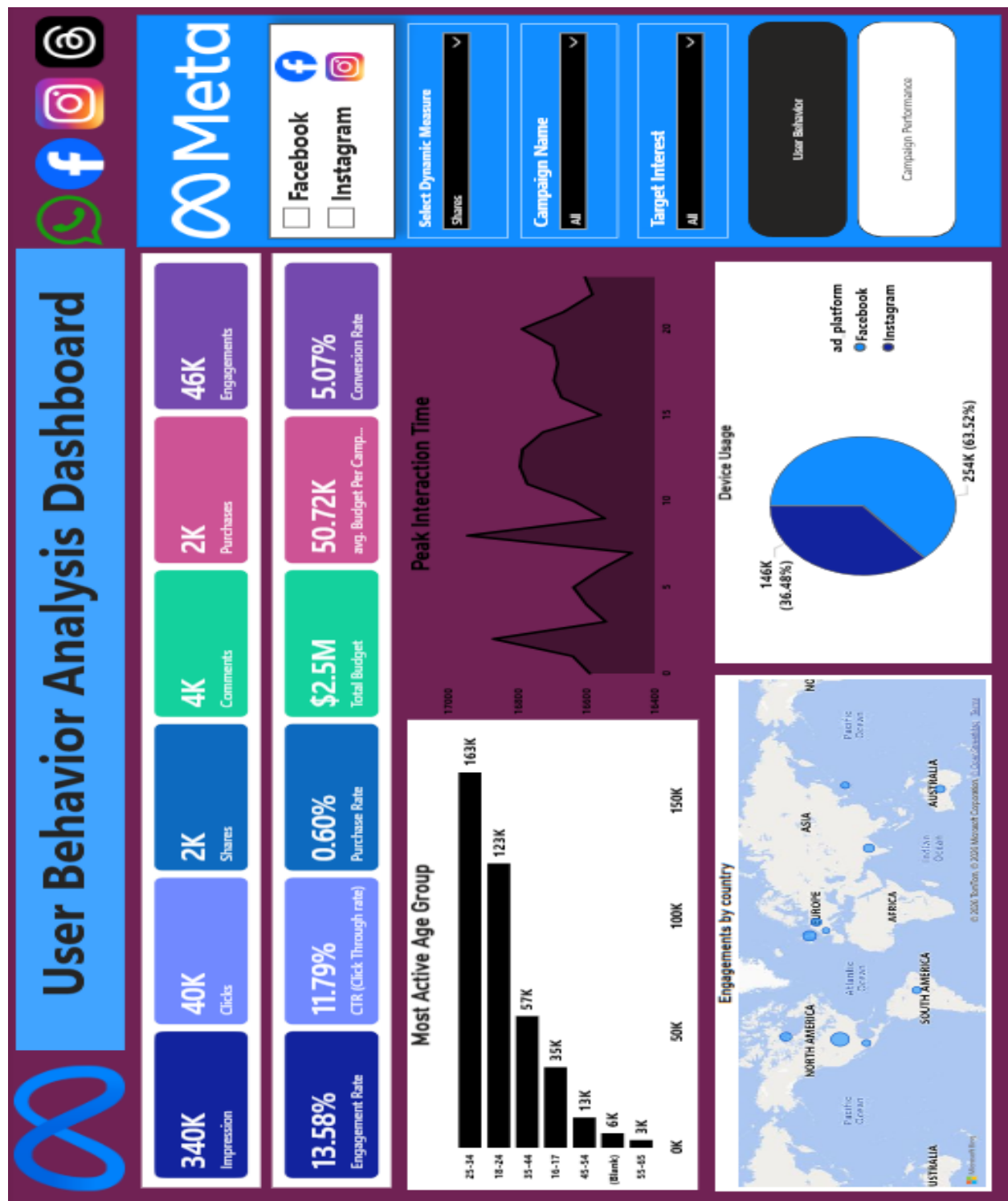


Figure 1.2

4.This dashboard shows ad campaign performance on Meta platforms. It tracks clicks, engagement, and conversions. It helps find the best campaigns and improve results.

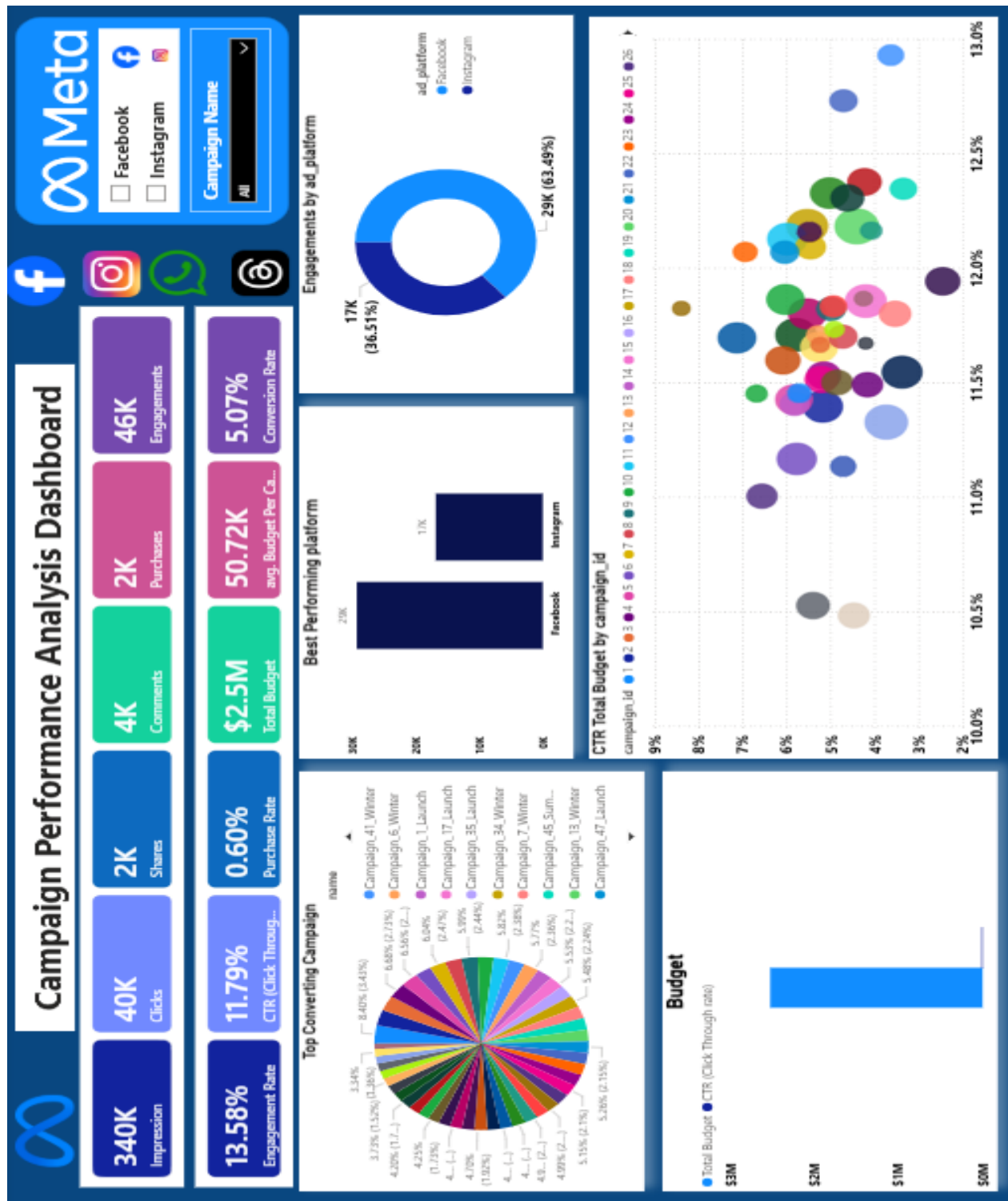


Figure 1.3

4. SOFT SKILLS DEVELOPMENT

During my internship, I not only developed technical expertise in data analytics and dashboard creation but also significantly enhanced my soft skills. These skills played a crucial role in improving my overall performance and helped me work more efficiently in a professional environment.

4.1 Communication Skills

I demonstrated strong communication skills throughout the internship by clearly expressing my ideas, asking relevant questions, and effectively interacting with mentors and peers. I was able to explain complex data insights in a simple and structured manner, which helped in better understanding during discussions and review sessions.

4.2 Teamwork & Collaboration

I actively participated in group discussions and collaborated with peers while working on different aspects of the project. I contributed my ideas confidently and also respected others' perspectives, which helped in creating a positive and productive team environment.

4.3 Problem-Solving Skills

While developing the Meta Ads Performance Dashboard, I encountered challenges such as data inconsistencies and errors in calculations. I approached these issues analytically, identified root causes, and implemented effective solutions, which improved the accuracy and quality of the dashboard.

4.4 Time Management

I effectively managed my time by completing assigned tasks within deadlines and maintaining consistency throughout the internship. I followed a structured approach using the log sheet, which helped me stay organized and track my daily progress.

4.5 Presentation Skills

I demonstrated strong presentation skills by confidently presenting my dashboard and explaining key insights during review sessions. I was able to communicate findings clearly, maintain good confidence, and engage the audience effectively. My ability to simplify technical concepts into understandable insights was one of my key strengths during the internship.

4.6 Professional Growth & Learning

This internship helped me build confidence, adaptability, and a professional mindset. I improved my ability to communicate effectively, present ideas clearly, and work in a structured manner. Overall, it contributed significantly to my personal and professional development.

5. CONCLUSION

During the internship, my primary focus was on designing and developing a comprehensive Meta Ads Performance Dashboard using data analytics and visualization techniques. This involved analyzing advertising data, identifying key performance metrics, and creating an interactive solution to effectively monitor and evaluate campaign performance.

The project included data cleaning and transformation using Power BI, creating calculated measures using DAX, and designing interactive dashboards with various visualizations. Key dashboards developed were user behavior analysis, campaign performance, audience insights, engagement quality, and overall, Meta Ads performance.

In addition to technical skills, this internship also helped me enhance my soft skills, particularly in communication and presentation. I was able to clearly explain my dashboard insights, participate in discussions, and confidently present my work during review sessions. This experience improved my ability to convey complex information in a simple and structured manner.

Overall, the internship provided a valuable learning experience by strengthening both my technical expertise and professional skills. It has enhanced my analytical thinking, problem-solving abilities, and confidence, preparing me for future roles in data analytics and business intelligence.

Key Insights

Technical Learning

Through this internship, I significantly enhanced my knowledge in data analytics and visualization. I gained hands-on experience in Power BI, data modeling, and DAX calculations. I learned how to create interactive dashboards, calculate important performance metrics such as Click-Through Rate (CTR) and Conversion Rate, and present data in a structured and meaningful way for better decision-making.

Challenges Faced

While working on the project, I encountered challenges such as handling missing data, maintaining data consistency, and ensuring accuracy in calculations. Selecting the most appropriate visualizations to represent complex data also required careful thinking and experimentation.

Overcoming these challenges helped me improve my analytical thinking and problem-solving skills.

Real-World Application

The Meta Ads Performance Dashboard has practical significance in the field of digital marketing. It helps businesses track campaign performance, understand user engagement, and make data-driven decisions. The dashboard provides clear insights that can be used to optimize marketing strategies and improve overall advertising effectiveness.

Learning Outcome

This internship provided complete exposure to the data analytics process, from data preparation to visualization and insight generation. In addition to technical skills, I also improved my soft skills, particularly in communication, teamwork, and presentation. I developed the ability to clearly explain complex insights, actively participate in discussions, and confidently present my work during review sessions.

Overall, the internship served as a valuable learning experience that strengthened both my technical expertise and professional skills. It enhanced my analytical thinking, problem-solving abilities, and confidence, preparing me for future opportunities in data analytics and business intelligence.

6. BIBLIOGRAPHY

The following references were used during the development of the Meta Ads Performance Dashboard project:

- **Microsoft Power BI Documentation:**

<https://learn.microsoft.com/en-us/power-bi/>

- **DAX Reference (Microsoft):**

<https://learn.microsoft.com/en-us/dax/>

- **Microsoft Excel Support:**

<https://support.microsoft.com/excel>

- **W3Schools (Data & Web Concepts):**

<https://www.w3schools.com/>

- **Google Scholar (Research References):**

<https://scholar.google.com/>

- **GitHub (Project References & Examples):**

<https://github.com/>

LOG SHEET

Sl. No	Date	Concept Learnt	Duration (in Hours)
1	23/02/2026	Excel Basics: Introduction & Interface	2 Hours
2	24/02/2026	Excel Formatting & Basic Formulas	2 Hours
3	25/02/2026	IF, COUNT, SUMIF	2 Hours
4	26/02/2026	Sorting, Filtering, Charts	2 Hours
5	27/02/2026	VLOOKUP & HLOOKUP	2 Hours
6	28/02/2026	INDEX & MATCH	2 Hours
7	03/03/2026	Nested IF & Logical Functions	2 Hours
8	04/03/2026	SUMIFS / COUNTIFS	2 Hours
9	05/03/2026	Pivot Tables	2 Hours
10	06/03/2026	Advanced Pivot & Slicers	2 Hours
11	09/03/2026	Data Validation	3 Hours
12	10/03/2026	Power Query Basics	3 Hours
13	11/03/2026	Excel Dashboard Creation	3 Hours
14	12/03/2026	Excel Mini Project	3 Hours
15	13/03/2026	Power BI Introduction	3 Hours
16	16/03/2026	Data Loading & Transformation	3 Hours
17	17/03/2026	Data Modeling	3 Hours
18	18/03/2026	Basic Visualizations	3 Hours

19	19/03/2026	Advanced Visualizations	3 Hours
20	20/03/2026	DAX Basics	3 Hours
21	21/03/2026	KPI Creation (CTR, Conversion Rate)	3 Hours
22	23/03/2026	Campaign Performance Dashboard	3 Hours
23	24/03/2026	User Behavior Dashboard	3 Hours
24	25/03/2026	Audience Insights Dashboard	3 Hours
25	26/03/2026	Engagement Quality Dashboard	3 Hours
26	29/03/2026	Meta Ads Performance Dashboard	4 Hours
27	30/03/2026	Slicers & Filters	3 Hours
28	31/03/2026	Dashboard Design & Formatting	3 Hours
29	01/04/2026	Drill-through & Interactions Report Navigation	3 Hours
30	02/04/2026	Python Basics: Variables & Data Types	3 Hours
31	03/04/2026	Conditional Statements & Loops	3 Hours
32	04/04/2026	Functions & Lists	3 Hours
33	05/04/2026	Dictionaries & Tuples	3 Hours
34	06/04/2026	NumPy Basics	3 Hours
35	7/04/2026	Pandas Introduction	3 Hours
36	8/04/2026	Data Frames & Series	3 Hours
37	9/04/2026	Data Cleaning & Missing Values	3 Hours
38	10/04/2026	Exploratory Data Analysis	3 Hours
39	11/04/2026	Group By & Aggregations	3 Hours

40	12/04/2026	Pivot Tables in Pandas	3 Hours
41	13/04/2026	Matplotlib Basics	3 Hours
42	14/04/2026	Advanced Plots (Bar, Line, Pie)	3 Hours
43	15/04/2026	SQL Basics: SELECT, WHERE	3 Hours
44	16/04/2026	GROUP BY, HAVING	3 Hours
45	17/04/2026	Aggregate Functions	3 Hours
46	18/04/2026	SQL JOINS	3 Hours
47	19/04/2026	Data Integration (Excel + SQL + Power BI)	3 Hours
48	20/04/2026	Dashboard Optimization	3 Hours
49	21/04/2026	Final Dashboard Improvements	3 Hours
50	22/04/2026	Report Documentation	3 Hours
51	23/04/2026	Final Project Presentation	3 Hours
52	24/04/2026	Internship Completion Summary	3 Hours

Total Duration: 150 hours

SOFT SKILLS

1	24/02/2026	Self-Awareness & Growth Mindset	3 Hours
2	25/02/2026	Fundamentals of Communication Skills	3 Hours
3	27/02/2026	Active Listening & Empathy Building	3 Hours
4	03/03/2026	Verbal & Non-Verbal Communication	3 Hours
5	05/03/2026	Confidence Building & Overcoming Fear	3 Hours
6	12/03/2026	Professional Etiquette & First Impressions	3 Hours

7	16/03/2026	Time Management & Productivity Techniques	3 Hours
8	20/03/2026	Critical Thinking & Decision Making	3 Hours
9	21/03/2026	Problem-Solving Strategies	3 Hours
10	23/03/2026	Teamwork & Collaboration Skills	3 Hours
11	5/04/2026	Adaptability & Handling Change	3 Hours
12	12/04/2026	Responsibility & Workplace Accountability	3 Hours
13	15/04/2026	Resume Building & Personal Branding	4 Hours
14	16/04/2026	Advanced Presentation Skills	4 Hours
15	20/04/2026	Professional Communication	4 Hours
16	21/04/2026	Interview Skills & HR Expectations	4 Hours
17	22/04/2026	Group Discussion & Public Speaking Lab	3 Hours
18	23/04/2026	Mock interview	3 Hours
19	24/04/2026	Final Assessment & Feedback	2 Hours
Total Duration: 60 Hours			

Declaration: It is declared that the student is completed her internship in our organization as per above schedule.

Signature of the Authority